

Improving Downstream Task Performance in Multilingual LLMs by Intervening Language Specific Neurons

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Introduction

Problem Statement

● Objective:

- Investigate the representation of language-specific information in large language models (LLMs).
- Develop methods to identify and utilize language-specific neurons for multilingual tasks.
- Enhance performance for low-resource languages by employing techniques such as activation probability entropy and LoRA-based fine-tuning.

● Importance:

- Ensure fair and inclusive NLP systems by bridging the performance gap between high-resource and low-resource languages.
- Advance understanding of the internal mechanisms of multilingual LLMs.
- Contribute to the development of efficient methods for improving multilingual performance.

Presentation Outline

- Review of existing methods for identifying and utilizing language-specific neurons.
- Experimental design: fine-tuning configurations and intervention strategies.
- Results and analysis for multilingual tasks (e.g., XNLI task).
- Conclusion and future work directions.

Methodology: Definition of Neuron

What is a Neuron?

- A **neuron** corresponds to a specific dimension of the activation vector $\mathbf{h}^i$, represented as:

$$\mathbf{h}^i = [\text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \otimes (\hat{\mathbf{H}}^i \mathbf{W}_3^i)] \cdot \mathbf{W}_2^i, \quad (1)$$

- The average activation (against Wikipedia sentence) of a neuron (i, j) across token positions is computed as:

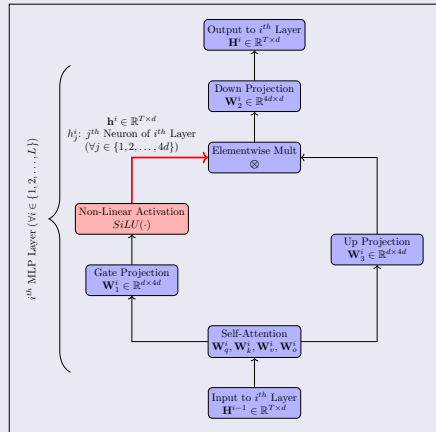
$$\bar{h}_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t. \quad (2)$$

- Language-specific activations are computed as:

$$h_{i,j}^l = \frac{1}{|S_l|} \sum_{s \in S_l} \bar{h}_{i,j}^{(s)}, \quad (3)$$

where S_l is the set of all sequences for language l .

MLP Architecture of Llama Model



Methodology: LAPE

LAPE Neurons (Tang et. al., 2024)

- To identify neurons with significant language-specific behavior, a threshold (for Wikipedia dataset) is defined as:

$$\tau = \text{Quantile}_{0.95} \left(p_{i,j}^l \mid \forall i, j, l \right). \quad (4)$$

- A neuron (i, j) is considered as language neuron if:

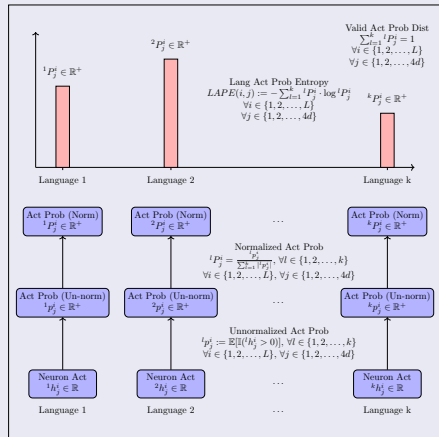
$$\exists l \in \{1, \dots, k\} \text{ such that } p_{i,j}^l > \tau \quad (5)$$

- Only 1% ($m = 0.01 \times 4d \times L$) of neurons with lowest LAPE values are selected for as language neurons.

$$LN = \text{argsort}_m \left\{ \text{LAPE}(i, j) \mid \forall i, j, \exists l \text{ s.t. } p_{i,j}^l > \tau \right\} \quad (6)$$

$$LSN(i, j) = \left\{ l \mid p_{i,j}^l > \tau \wedge (i, j) \in LN \right\} \quad (7)$$

LAPE Workflow



Methodology: Activation Based Methods

Neuron Relevance

- **Activation Probability 90p:** Measures the probability of a neuron's activation exceeding the 90th percentile value for its layer.
- Relevance for a neuron (i, j) in language l is computed from its activation against Wikipedia dataset:

$$P_q(h_{i,j}^l) = \inf \left\{ x \in \mathbb{R} : CDF_{h_{i,j}^l}(x) \geq q \right\} \quad (8)$$

$$r_{i,j}^l = \mathbb{E} \left[\mathbb{I} \left(h_{i,j}^l > P_{0.9}(h_{i,j}^l) \right) \right] \quad (9)$$

- Aggregates relevance values across neurons:

$$R^l = \{r_{i,j}^l \mid i \in \{1, \dots, L\}, j \in \{1, \dots, 4d\}\}. \quad (10)$$

Activation Based Neurons

- **Selection Process:** Identify top neurons with the highest relevance for each language.
- Select top 0.125% neurons per language based on their maximum relevance:

$$m = \frac{0.125 \times L \times 4d}{100} \quad (11)$$

$$N^l = \text{argsort_desc}_m \left\{ r_{i,j}^l \mid \forall i, j \right\} \quad (12)$$

- This process ensures that neurons with the highest language-specific relevance are identified for each language.

Methodology: Neuron Fine-Tuning using LoRA

LoRA Fine-Tuning for a Task

- **LoRA Mechanism:** Fine-tunes a subset of parameters in neural networks by introducing a low-rank decomposition:

$$[\text{masked_A}]_{ij} = 1, \forall i \in \{1, \dots, d\}, \forall j \in \{1, \dots, r\} \quad (13)$$

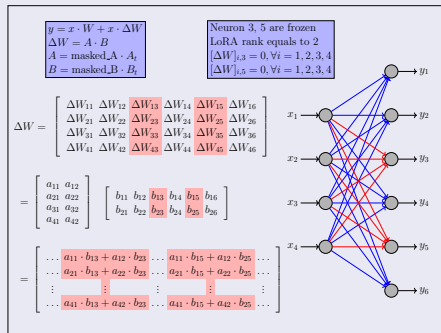
$$[\text{masked_B}]_{ij} = \begin{cases} 0, & \text{if } j \in \text{Frozen Neurons,} \\ 1, & \text{otherwise.} \end{cases} \quad (14)$$

- To freeze specific neurons, columns of B corresponding to the frozen neurons are masked with zeros. This prevents any updates to these neurons during fine-tuning.

$$\Delta W_{i,j} = 0, \quad \forall i \in 1, \dots, d, \forall j \in \text{Frozen Neurons} \quad (15)$$

- LoRA fine-tunes the MLP layer for the trainable language neurons and attention layer (q, k, v, o projections). The task-specific classifier head is fully trained.

Targeted LoRA Fine-Tuning Example



Experiments: Dataset and Baseline

Task and Dataset: XNLI

- **XNLI Task:** Multilingual natural language inference task to classify relationships between premise and hypothesis into *entailment*, *neutral*, or *contradiction*.
- **Dataset:** Sentence pairs annotated across 15 languages. Focused on English (*en*) as source language and Vietnamese (*vi*), Hindi (*hi*), and Urdu (*ur*) as target languages.
- For the experiments, a fraction $\mathcal{F}$ is used to train the model as $|\mathcal{D}_l^{\text{train}}| = 400K$.

$$|\mathcal{F}| = \text{frac} \times |\mathcal{D}_l^{\text{train}}| \quad (16)$$

where $\text{frac} = 0.25$ is the fraction of the training dataset used.

- **Objective:** Improving the zero-shot performance for low-resource languages (*vi*, *hi*, *ur*).

Baseline Experiments (No Eval Time Interventions!)

- **Zero-Shot Baseline:** Fine-tune on English (*en*) XNLI dataset. Evaluate zero-shot transfer to target languages (*vi*, *hi*, *ur*):

$$\text{Accuracy} = \frac{\sum_{i=1}^N \mathbb{I}(\hat{y}^i = y^i)}{N} \quad (17)$$

where N is the total test examples.

- **Direct Fine-Tuning Baseline:** Fine-tune on target language such as Vietnamese (*vi*) XNLI dataset to establish the upper bound for *vi* performance. Accuracy remains the evaluation metric.

Experiments: Goals and Fine-Tuning Strategy

Experiment Goals

- **Objective:** Improve zero-shot performance on target language (vi) without using target language training data.
- **Constraints:**
 - No target language training data (vi) allowed.
 - Only English (en) XNLI training data available.
 - Use model surgery techniques (e.g., neuron fine-tuning, activation pattern interventions).

- **Target Metric:**

$$\Delta_{ZS} = \text{Acc}(\mathcal{M}_{\text{experiment}}) - \text{Acc}(\mathcal{M}_{\text{en}}) \quad (18)$$

- Maximize Δ_{ZS} to close the performance gap between zero-shot and direct fine-tuning.

Fine-Tuning Setups

- **Setup A:** Standard fine-tuning using LoRA applied to q, k, v, o projection matrices and the classification head. Evaluated with statistical activation substitutions ($\mu, 90p, 95p$, etc.).
- **Setup B:** Fine-tuning with LoRA targeting English-specific neurons ($\mathcal{N}_{\text{en}}$) in the MLP block.
- **Setup C:** Fine-tuning with LoRA targeting Vietnamese-specific neurons ($\mathcal{N}_{\text{vi}}$) in the MLP block.
- **Setup D:** Combined fine-tuning targeting both English ($\mathcal{N}_{\text{en}}$) and Vietnamese ($\mathcal{N}_{\text{vi}}$) neurons with LoRA.

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Thank You!